Patchwork Preservation under $\mathcal{ALCI}_{\mathsf{RCC5}}$ Augmentation

A probe toward undecidability

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Abstract

The decidability argument for $\mathcal{ALCI}_{\mathsf{RCC5}}$ rests on a specific corollary of the Renz–Nebel patchwork property for atomic RCC5 networks: that typed arc-consistency is complete for realizability in the presence of an $\mathcal{ALCI}$ TBox. If this *typed patchwork* fails—if there exists a TBox whose arc-consistent typed networks are not globally realizable—then undecidability could plausibly hide in the failure gap. This paper probes that gap. We formalize typed patchwork, separate it from atomic patchwork, attempt three candidate counterexamples, and report on what we find. Our finding is that typed patchwork *appears to hold* in each probe, with every apparent obstruction reducing either to local arc-inconsistency or to type-level unsatisfiability. We identify precisely where the analysis would need to break for undecidability to emerge, and argue this aligns structurally with the completeness-extraction paper's "realized relations are automatically relation-safe" key lemma. The probe is negative, but it sharpens the question: any future undecidability proof for $\mathcal{ALCI}_{\mathsf{RCC5}}$ must either exhibit a typed-patchwork failure of a specific combinatorial shape, or bypass patchwork entirely.

1 Introduction

Every decidability argument for $\mathcal{ALCI}_{\mathsf{RCC5}}$ in this project—the quasimodel method, the cover-tree tableau, the split-forest semantics, the completeness-extraction paper—ultimately appeals to Renz and Nebel's patchwork property [1] for the RCC5 calculus. The patchwork property says that a finite atomic RCC5 network is globally consistent iff it is path-consistent. It is the single load-bearing combinatorial fact behind everything we have claimed.

The question this paper asks is whether the patchwork property *survives* augmentation with an $\mathcal{ALCI}$ TBox. More precisely: does path-consistency of a typed RCC5 network—a network whose nodes carry Hintikka types and whose edges are locally type-safe—imply the existence of a global model satisfying the TBox? We call this the *typed patchwork* question.

If typed patchwork holds, the decidability argument for $\mathcal{ALCI}_{\mathsf{RCC5}}$ is robust to TBox pressure and the currently-claimed positive result stands. If typed patchwork *fails*, there is an opening for undecidability: the failure would mean that some typed network is locally consistent at every arc but has no global model, and one could then try to amplify such a failure into an undecidable reduction.

The reason this is the cleanest question to attack is that both outcomes are informative. A failure of typed patchwork does not automatically give undecidability (one would still need to encode computation into failure patterns), but it would invalidate every existing decidability proof in this repository. A confirmation of typed patchwork, conversely, explains why all the undecidability reductions documented in [6] are blocked and points future undecidability work toward the one remaining loophole: reductions that do not pass through any patchwork-style argument at all.

Contributions.

- We formalize three strengths of typed patchwork (§2, §3) and separate them from the untyped Renz–Nebel property.
- We attempt three candidate counterexamples (§4–§6) selected for structural diversity: forced coincidence of distinct types under shared parentage, inverse-role cycle closure, and deep witness chains with interleaved universals.
- We analyse each probe and report that each fails to produce a counterexample, with the obstruction reducing either to local arc-inconsistency or to type-level unsatisfiability (§7).
- We identify the precise structural shape a genuine typed-patchwork failure would need to take (§8), and argue it aligns with the dual of the completeness-extraction key lemma.
- We conclude with an honest assessment of what this probe does and does not establish (§10).

This paper is deliberately short and deliberately negative. It is a research probe, not a proof. We invite the reader to find the counterexample we could not.

2 Preliminaries

We briefly recap RCC5, Hintikka types, and the Renz–Nebel patchwork property. The reader is referred to [4, 2, 3] for full definitions.

RCC5. The five base relations are $\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}$, with EQ interpreted as identity (strong-EQ semantics) and $\mathsf{PP}, \mathsf{PPI}$ as converse partial orders. For distinct nodes, the four admissible atomic relations are $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$. The composition table comp is well known; we write $\mathrm{comp}(R_1, R_2)$ for the set of RCC5 relations consistent with the composition of R_1 and R_2.

Atomic networks and patchwork. An *atomic* RCC5 *network* is a pair (V, ρ) where V is finite and $\rho : V \times V \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ satisfies $\rho(x, x) = \mathsf{EQ}$, $\rho(y, x) = \mathrm{inv}(\rho(x, y))$, and

ρ is *path-consistent*:

$$\rho(x,z) \in \mathrm{comp}(\rho(x,y), \rho(y,z)) \quad \text{for all } x,y,z \in V.$$

Theorem 2.1 (Renz and Nebel [1])**.** *A finite atomic* RCC5 *network is realizable iff it is path-consistent.*

Hintikka types. Fix a concept C_0 and a TBox T; let $\mathrm{cl}(C_0, T)$ be the closure under subconcepts, negations, and TBox axioms. A *Hintikka type* (for (C_0, T)) is a subset $\tau \subseteq \mathrm{cl}(C_0, T)$ that is maximally locally consistent: for every Boolean subconcept it contains exactly one of $\{D, \neg D\}$, for each $D_1 \sqcap D_2$ it contains D_1 and D_2, for each $D_1 \sqcup D_2$ it contains at least one, and so on. We write $\mathrm{Tp}(C_0, T)$ for the finite set of Hintikka types.

Type-safety. Given an edge $\rho(x,y) = R$ with $\mathrm{tp}(x) = \alpha$, $\mathrm{tp}(y) = \beta$: the edge is *type-safe* if for every universal $\forall S.D \in \alpha$ with $S = R$, we have $D \in \beta$, and symmetrically for $\mathrm{inv}(R)$. Write $\mathrm{TypeSafe}(\alpha, \beta) \subseteq \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ for the set of RCC5 relations R with (α, R, β) type-safe.

3 Three strengths of typed patchwork

We now state three progressively stronger patchwork claims. The first is Renz–Nebel; the third is what the decidability proof needs.

Definition 3.1 (Atomic patchwork)**.** For every finite path-consistent atomic RCC5 network (V, ρ), there exists a model M of the RCC5 algebra and an injection $\iota : V \to M$ such that $\rho(x,y) = \rho^M(\iota(x), \iota(y))$ for all $x, y \in V$.

Theorem 3.2. *Atomic patchwork holds (Theorem 2.1).*

Definition 3.3 (Locally-coherent typed patchwork)**.** Fix an $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox T and concept C_0. A *typed network* is a triple (V, ρ, τ) with (V, ρ) a path-consistent atomic RCC5 network and $\tau : V \to \mathrm{Tp}(C_0, T)$ such that every edge is type-safe. Locally-coherent typed patchwork says: every typed network extends to a model $M \models T \cup \{C_0\}$ with an injection $\iota : V \to M$ preserving ρ and τ.

This is already stronger than Definition 3.1: it requires not merely an RCC5 model but a model that satisfies the TBox with the prescribed types. Additional existential witnesses must be supplied for every $\exists R.D \in \tau(x)$, and those witnesses must be globally consistent with the rest of V.

Definition 3.4 (Arc-consistent typed patchwork)**.** Given T, C_0 as above. A *disjunctive typed network* is (V, D, τ) where $D : V \times V \to 2^{\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}}$ assigns a non-empty set of admissible atomic relations to each ordered pair, subject to inv-symmetry. *Arc-consistency* is the fixed point of the propagation

$$D'(x,z) \;:=\; D(x,z) \cap \bigcup_{R_1 \in D(x,y)} \mathrm{comp}(R_1, D(y,z)) \cap \mathrm{TypeSafe}(\tau(x), \tau(z)).$$

Arc-consistent typed patchwork says: if (V, D, τ) is arc-consistent and every $D(x, y)$ is non-empty, then there exists a model $M \models T \cup \{C_0\}$ realizing some atomic refinement of (V, D, τ).

The split-forest semantics paper [2] establishes arc-consistent typed patchwork for a *restricted* class of disjunctive networks (those arising from valid finite rank-d quotients) via the LCA-induction proof of Theorem 1.17 plus König's lemma on canonical refinements. The completeness-extraction paper [3] establishes the converse: every model induces a quotient that satisfies arc-consistency, via the key lemma that realized relations are automatically relation-safe.

The gap. What is *not* established, to our knowledge, is Definition 3.4 for *arbitrary* arc-consistent typed networks—not just those arising from quotients of genuine models. This is where a counterexample could live. If there is an arc-consistent (V, D, τ) not realizable by any model of T, then arc-consistency alone is not complete, and the decidability argument needs further propagation.

We turn to three probes for such a counterexample.

4 Probe 1: forced coincidence of distinct types

The idea: force three nodes x, y, z with $x\mathsf{PP}y$ and $x\mathsf{PP}z$, and use the TBox to demand that y and z have types B and C that are pairwise incompatible on every non-EQ atomic relation. Renz–Nebel allows $y?z \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$; if the TBox forbids four of the five via TypeSafe, only EQ remains, and if $B \sqcap C$ is itself unsatisfiable as a type, the network is unrealizable.

Probe 4.1 (Forced-coincidence probe)**.** Let A, B, C be distinct concept names. Let

$$\begin{aligned} &A \sqsubseteq \exists\mathsf{PP}.B \sqcap \exists\mathsf{PP}.C, \\ &B \sqsubseteq \forall\mathsf{DR}.\neg C \sqcap \forall\mathsf{PO}.\neg C \sqcap \forall\mathsf{PP}.\neg C \sqcap \forall\mathsf{PPI}.\neg C, \\ &B \sqsubseteq \neg C. \end{aligned}$$

Take $V = \{x, y, z\}$ with $\tau(x) \supseteq \{A\}$, $\tau(y) \supseteq \{B\}$, $\tau(z) \supseteq \{C\}$, and $\rho(x, y) = \mathsf{PP}$, $\rho(x, z) = \mathsf{PP}$.

Observation 4.2. The network in Probe 4.1 is *not* arc-consistent.

Proof. The propagation $D(y, z)$ starts from $\mathrm{comp}(\mathsf{PPI}, \mathsf{PP}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ (path-consistency allows all five). Type-safety filters by $\tau(y) = B$: the universals in B rule out all non-EQ relations on the y-side, so $\mathrm{TypeSafe}(\mathrm{tp}(y), \mathrm{tp}(z)) \cap \mathrm{comp}(\mathsf{PPI}, \mathsf{PP}) \subseteq \{\mathsf{EQ}\}$. But $\rho(y, z) = \mathsf{EQ}$ forces $y = z$, hence $\mathrm{tp}(y) = \mathrm{tp}(z)$, hence $B \sqcap C \in \mathrm{tp}(y)$—but $B \sqsubseteq \neg C$, so $B \sqcap C$ is type-inconsistent and there is no Hintikka type containing it. Arc-consistency propagation therefore empties $D(y, z)$. □

The probe reduces to an *arc-inconsistency*. It is caught by the AC-3 loop and never presented to the global realizability question at all. This is exactly what one would hope: typed patchwork is not even tested here, because the local filter already fails.

The probe is structurally isomorphic to the "coincidence obstruction" of Wessel 2002 [5, Fig. 11] *except that in Wessel's setting,* EQ *is not ruled out by a universal*; there, the obstruction

is that nothing forces the cells to be distinct, so the grid cells silently identify. In our probe, the TBox *forces* the cells to be distinct via $B \sqsubseteq \neg C$, but then arc-consistency catches the conflict locally. The two obstructions are dual.

5 Probe 2: inverse-role cycle closure

The idea: use inverse roles to force a cycle $x \to y \to z \to x$ of role compositions, and hope that the composition table permits the cycle locally at each arc but no global atomic assignment closes it consistently with the TBox.

Probe 5.1 (Inverse-cycle probe)**.** Let A, B, C, D be concept names. Consider

$$\begin{aligned} A &\sqsubseteq \exists \mathsf{PP}.B, \\ B &\sqsubseteq \exists \mathsf{PO}.C, \\ C &\sqsubseteq \exists \mathsf{PPI}.D, \\ D &\sqsubseteq \forall \mathsf{PP}.\neg A. \end{aligned}$$

Take $V = \{x, y, z, w\}$ with $\tau(x) \supseteq \{A\}$, $\tau(y) \supseteq \{B\}$, $\tau(z) \supseteq \{C\}$, $\tau(w) \supseteq \{D\}$, and edges $\rho(x, y) = \mathsf{PP}$, $\rho(y, z) = \mathsf{PO}$, $\rho(z, w) = \mathsf{PPI}$. The fourth edge $\rho(x, w)$ is unconstrained by explicit existentials.

Observation 5.2. The network in Probe 5.1 is arc-consistent and realizable.

Proof. Compose the path $x \to y \to z \to w$: $\mathrm{comp}(\mathsf{PP}, \mathsf{PO}) \ni \mathsf{PP}, \mathsf{PO}, \mathsf{DR}$; then with PPI on the z-w edge, we get $\rho(x, w) \in \mathrm{comp}(\mathrm{comp}(\mathsf{PP}, \mathsf{PO}), \mathsf{PPI})$, which is all of $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ (the composition of $\{\mathsf{PP}, \mathsf{PO}, \mathsf{DR}\}$ with PPI covers every relation). Type-safety from $D \sqsubseteq \forall \mathsf{PP}.\neg A$ rules out $\rho(w, x) = \mathsf{PP}$, i.e., $\rho(x, w) = \mathsf{PPI}$. Remaining candidates: $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{EQ}\}$. EQ is ruled out if $A \sqcap D$ is type-inconsistent (it is, via the universal on D reflecting back: if $w = x$ then $w\mathsf{PP}\cdot$ but $x\mathsf{PP}y$ with $\tau(y) = B \neq \neg A$—wait, this doesn't immediately fire unless A is in $\tau(y)$, which it is not). Actually EQ merely forces $\tau(x) = \tau(w)$, hence $\{A, D\} \subseteq \tau$. This is type-consistent only if $A \sqcap D$ is satisfiable, which is not ruled out by the TBox. We pick $\rho(x, w) = \mathsf{DR}$ for a concrete realization: a model with four distinct points and the specified relations satisfies all axioms. □

The probe produces an arc-consistent, realizable network. No obstruction.

Variants. We attempted several variants: longer cycles (5, 6 nodes), cycles with EQ-forcing universals, cycles that interleave PO and PP in ways designed to exhaust the composition table's slack. Every variant either (a) produced a realizable network, or (b) collapsed to arc-inconsistency via type-safety filtering on some edge. We did not find a variant that produced a path-consistent, arc-consistent, type-coherent network without a model.

6 Probe 3: deep witness chains with interleaved universals

The idea: an $\mathcal{ALC}$-style "yardstick" construction. Use a long chain of existential witnesses with universals that accumulate along the chain, forcing an obligation that cannot be satisfied at the chain's end without backtracking.

Probe 6.1 (Yardstick probe)**.** Fix $n \geq 2$ and concept names $A_0, A_1, \ldots, A_n, F$. Let

$$\begin{aligned} A_i &\sqsubseteq \exists \mathsf{PP}.A_{i+1} \quad (0 \leq i < n), \\ A_i &\sqsubseteq \forall \mathsf{PP}.F \quad (0 \leq i \leq n-1), \\ A_n &\sqsubseteq \neg F. \end{aligned}$$

Observation 6.2. Probe 6.1 is type-unsatisfiable for every $n \geq 1$.

Proof. Every model of A_0 has a PP-successor of type A_1; by the universal $\forall \mathsf{PP}.F \in A_0$, this successor also has F. But A_1 contains $\forall \mathsf{PP}.F$ and (for $n \geq 2$) by iteration A_{n-1} produces A_n with F, contradicting $A_n \sqsubseteq \neg F$. So there is no Hintikka type containing A_0. □

Refining the probe. The above probe is caught by type-saturation before any network is constructed. A more interesting variant replaces the universal with something that only fires conditionally on another concept that might be absent at the chain's end. We tried:

$$\begin{aligned} A_i &\sqsubseteq \exists \mathsf{PP}.A_{i+1}, \\ A_i &\sqsubseteq \forall \mathsf{PP}.(G \sqcup \neg G), \\ A_n &\sqsubseteq \neg G \sqcap \neg(\neg G). \end{aligned}$$

Of course $\forall \mathsf{PP}.(G \sqcup \neg G)$ is trivially true and the bottom constraint $A_n \sqsubseteq \neg G \sqcap G$ is vacuously unsatisfiable in types. After several variants, the failure mode was uniform: either the type closure caught an inconsistency immediately, or the type was satisfiable and a trivial model existed.

The underlying reason: $\mathcal{ALC}$ universals propagate deterministically along role chains. In $\mathcal{ALCI}$, inverse roles are also deterministic. In $\mathcal{ALCI}_{\mathsf{RCC5}}$ on top, the role labels are further constrained by $\mathsf{RCC5}$ composition via arc-consistency, but composition only *loosens* the constraint (unions of allowed relations), never tightens it below what the types already force. The yardstick construction is therefore flattened by arc-consistency rather than amplified.

7 Structural analysis: why every probe fails

Across all three probes (and several unreported variants), the obstruction to realizability reduced to one of two phenomena:

1. **Local arc-inconsistency:** $\mathrm{TypeSafe}(\tau(x), \tau(y)) \cap \mathrm{comp}(\ldots) = \varnothing$ for some pair, and AC-3 empties $D(x, y)$.
2. **Type-level unsatisfiability:** $\bigcap_{\forall R.D_i \text{ firing at } \tau(x)} \{D_i\}$ is inconsistent, and the Hintikka-type closure catches it.

Neither of these is a failure of typed patchwork in the sense of Definition 3.4, because both occur *before* arc-consistency terminates with all $D(x, y)$ non-empty.

A structural reason. The completeness-extraction paper [3] identifies the key lemma: in a genuine model M, every realized atomic relation between two elements $x, y \in M$ is automatically relation-safe for $(\mathrm{tp}^M(x), \mathrm{tp}^M(y))$. The dual statement is: if no atomic relation is relation-safe for (α, β) under composition $\mathrm{comp}(R_1, R_2)$, then no model can realize α and β at those positions.

Typed patchwork is essentially the contrapositive: *if* every arc of a typed network has a non-empty filtered domain, *then* some model realizes the network. The probes fail to produce a counterexample because, intuitively, non-empty filtered domains *are* the witness that a model exists.

Proposition 7.1 (Conjectural; we were unable to falsify)**.** *For every $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox T and satisfiable concept C_0, typed patchwork holds in the sense of Definition 3.4: every arc-consistent typed network with non-empty filtered domains is realizable by some model of $T \cup \{C_0\}$.*

A full proof of Proposition 7.1 would amount to a simpler alternative to the split-forest construction: extend the network by Henkin witnesses, maintaining arc-consistency at each step. The step that is *not obviously correct* is that adding a new witness for some existential $\exists R.D$ at a node x requires choosing $\rho(y, z)$ for every existing $z \in V$, and these choices must be simultaneously arc-consistent. The split-forest paper handles this via canonical refinements and König's lemma on an infinite tree of choices. We do not prove Proposition 7.1 here; we merely note that our probes were consistent with it.

8 The shape of a genuine failure

Suppose, contrary to Proposition 7.1, that typed patchwork fails. What would the counterexample look like?

1. It would be an arc-consistent typed network with all $D(x, y) \neq \varnothing$.
2. The model-realizability question would fail *not* at any individual arc (all arcs are locally consistent), but at a *global* level where the simultaneous atomic choices cannot be made.
3. The failure must survive arc-consistency. Since AC-3 propagates through ternary composition, the failure mechanism must be essentially k-ary for some $k \geq 4$: four or more nodes whose pairwise filtered domains are all non-empty but whose joint atomic assignments are all inconsistent with the TBox.

Why this shape is hard to achieve. Renz–Nebel Theorem 2.1 already forecloses the RCC5-only version of such k-ary failures: atomic path-consistency is complete for realizability of untyped networks. To manufacture a k-ary typed failure, one would need the TBox to introduce a k-ary constraint that is *invisible to any triple of nodes* but *visible to the full k-subset.*

In $\mathcal{ALCI}$, every constraint is expressible as a restriction on a single node's type (universals, existentials, Boolean combinations). The TBox does not directly express k-ary constraints

for $k \geq 2$. TBox axioms become node-local (unary) constraints after Hintikka closure. Arc constraints ($k = 2$) are captured by type-safety. Triple constraints ($k = 3$) are captured by path-consistency / composition. *There is no $\mathcal{ALCI}$ mechanism for introducing a genuinely k-ary constraint for $k \geq 4$.*

This is the structural reason we conjecture typed patchwork holds. Any k-ary-looking constraint in $\mathcal{ALCI}$ decomposes into unary, binary (type-safety), or ternary (composition) pieces, all of which are captured by arc-consistency.

What would break this reasoning. The only loophole we see is if *inverse roles plus composition* could *synthesize* a k-ary constraint through a long chain. The canonical example is $\exists R.\forall R^{-}.A$: this uses both directions of a single role to express "every R-predecessor of some R-successor is A." In principle, iterating such constructions could accumulate global constraints that no finite arc-consistency check sees.

However, in the $\mathcal{ALCI}_{\mathsf{RCC5}}$ context, role inverse is not independent of $\mathsf{RCC5}$ composition: $\mathsf{RCC5}$ relations are their own inverse under inv, and composition is associative. Long chains of $\exists R.\forall R^{-}\ldots.$ collapse to unary constraints under type closure. So we do not see a route to k-ary synthesis via inverse role alone.

If such a route exists, the counterexample to Proposition 7.1 would be the vehicle to exhibit it, and it would simultaneously be a counterexample to the decidability proof. This is the single tightest place to look for undecidability.

9 Implications for undecidability

If typed patchwork holds (Proposition 7.1):

- The decidability proof for $\mathcal{ALCI}_{\mathsf{RCC5}}$ is robust.
- Every standard undecidability reduction (grid encodings, PCP, word problems) is blocked by the same mechanism: such reductions require global rigidity, which typed arc-consistency does not permit.
- A non-trivial undecidability proof would need to bypass patchwork entirely—encode computation into something other than the relational structure. Given that $\mathcal{ALCI}_{\mathsf{RCC5}}$'s only semantic content *is* the relational structure, such a bypass seems unavailable.

If typed patchwork fails:

- The failure exhibits a concrete k-ary constraint ($k \geq 4$) that arc-consistency misses.
- One would then need to amplify the failure into a computation-simulating reduction. This is non-trivial but not impossible: a single k-ary failure at fixed k does not imply undecidability, but a *family* of failures indexed by the tape content of a Turing machine would.

We emphasize: *finding a typed-patchwork failure does not by itself prove undecidability.* It only opens the door. But *not* finding one—as in this paper—closes the most obvious door we had.

10 Honest assessment

This paper is a negative result in both directions. We did not find a counterexample, and we did not prove Proposition 7.1. What we did establish:

1. The typed patchwork question can be precisely formalised (Definition 3.4) as a question about arc-consistency completeness.
2. Several natural candidate counterexamples (forced coincidence, inverse cycle, yardstick) reduce to arc-inconsistency or type-inconsistency before reaching the global realizability question.
3. The structural reason for this reduction is that $\mathcal{ALCI}$ cannot express k-ary constraints for $k \geq 4$ that are invisible to arc-consistency.

What we did *not* establish:

1. A full proof of typed patchwork. This would require Henkin-style model construction maintaining arc-consistency, essentially a repackaging of the split-forest soundness direction.
2. A sharp characterisation of which $\mathcal{ALCI}$ constructions could in principle synthesise k-ary constraints through inverse-role interaction. We sketched an argument that none do, but did not prove it rigorously.
3. An exhaustive search of candidate counterexamples. Three probes and a handful of variants is not exhaustive; a reader with a better combinatorial imagination may well find one.

Outlook. The most promising refinement of this probe would be to formalize the "no k-ary synthesis" claim and prove it rigorously. A theorem of the form

> *Every $\mathcal{ALCI}_{\mathsf{RCC5}}$ constraint on a finite network decomposes into constraints of arity at most three, and arc-consistency is complete for all arity-≤ 3 constraint satisfaction problems over* $\mathsf{RCC5}$.

would close both questions simultaneously: it would prove typed patchwork and simultaneously certify that no $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox can express an undecidable CSP.

A theorem of that shape would provide an alternative, more direct decidability proof than the split-forest construction, grounded in constraint satisfaction rather than tree semantics. It would also finally give a *principled* answer to why every undecidability reduction fails: not a survey of blocked techniques, but a single theorem that all are instances of.

We leave this to future work.

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